

Friedel–Crafts Alkylation Catalyzed by the Artificial Enzyme LmrR_pAF

1 Introduction

Artificial enzymes that incorporate unnatural amino acids provide a powerful strategy for introducing novel catalytic functions into protein scaffolds. One such system is the transcriptional regulator LmrR modified with the unnatural amino acid *p*-aminophenylalanine (pAF). The structure of this artificial enzyme has been solved and deposited in the Protein Data Bank under accession code 6I8N. In this system, the primary amine of pAF acts as an organocatalytic group capable of forming iminium intermediates with aldehyde substrates. This mechanism enables the enzyme to catalyze Friedel–Crafts alkylation reactions between α , β -unsaturated aldehydes and indole nucleophiles. The hydrophobic pocket formed at the LmrR dimer interface provides a confined environment that helps orient the substrates and stabilize reactive intermediates during catalysis.

2 Overall Catalytic Cycle

The catalytic cycle consists of three major steps:

1. Formation of an iminium intermediate between pAF and the aldehyde substrate
2. Carbon–carbon bond formation through a Friedel–Crafts reaction
3. Hydrolysis of the iminium intermediate and regeneration of the catalyst

These steps collectively enable the formation of a new C–C bond between the aldehyde substrate and the indole nucleophile.

3 Reaction 1: Formation of the Iminium Intermediate

The catalytic cycle begins with nucleophilic attack of the primary amine of the *p*-aminophenylalanine (pAF) residue on the carbonyl carbon of the α , β -unsaturated aldehyde substrate. In the free aldehyde, the carbonyl carbon is sp^2 hybridized and adopts a trigonal planar geometry with bond angles close to 120° .

When the lone pair of the pAF amine attacks the carbonyl carbon, the π bond of the carbonyl group is broken and the oxygen atom becomes negatively charged. This nucleophilic addition converts the carbonyl carbon from an sp^2 hybridized planar center to an sp^3 hybridized tetrahedral center, producing a carbinolamine intermediate.

The resulting tetrahedral intermediate contains four substituents around the former carbonyl carbon: the hydroxyl group, the nitrogen of pAF, the aldehyde substituent, and the hydrogen originally attached to the carbonyl carbon. Proton transfer steps then occur within the active site, allowing the hydroxyl group to be protonated and subsequently eliminated as water.

Loss of water restores conjugation and generates an iminium ion. In the iminium intermediate, the carbon bound to nitrogen returns to an sp^2 hybridized state and adopts a trigonal planar geometry. The positive charge is delocalized over the C=N bond and the adjacent conjugated system, which significantly increases the electrophilicity of the β -carbon of the substrate.

4 Reaction 2: Friedel–Crafts Carbon–Carbon Bond Formation

The activated iminium intermediate then undergoes nucleophilic attack by the indole substrate. The most nucleophilic position of indole is the C3 carbon, which donates π electrons to the β -carbon of the iminium-activated enal system.

Because the iminium carbon is sp^2 hybridized and planar, the nucleophile can approach from either face of the double bond. Within the enzyme pocket, however, steric constraints imposed by the LmrR scaffold orient the substrates such that the indole approaches along a trajectory that favors formation of the new carbon–carbon bond.

During this step, the π electrons from the indole ring form a new σ bond with the β -carbon, temporarily generating a positively charged Wheland-type intermediate on the indole ring. Deprotonation of the indole restores aromaticity and produces an alkylated iminium intermediate that remains covalently bound to the pAF residue.

5 Reaction 3: Hydrolysis and Catalyst Regeneration

In the final stage of the catalytic cycle, a water molecule acts as a nucleophile and attacks the electrophilic carbon of the iminium intermediate. This step again converts the iminium carbon from an sp^2 planar geometry to an sp^3 tetrahedral geometry, generating a second carbinolamine intermediate.

Subsequent cleavage of the carbon–nitrogen bond releases the Friedel–Crafts alkylated product and regenerates the free amine of the pAF residue. The catalytic residue therefore returns to its original state and is capable of initiating another catalytic cycle.

6 Modeling the Mechanism Using PyMOL

The reaction mechanism can be modeled using the PyMOL molecular visualization program using the following workflow:

1. Download the crystal structure with PDB ID 6I8N.
2. Load the structure into PyMOL.
3. Remove crystallization components if necessary:

remove resn MES

4. Identify the catalytic residue:

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select paf, resi 15  
show sticks, paf
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5. Position the α, β -unsaturated aldehyde substrate near the pAF amine with an approximate distance of 3 Å between the nitrogen atom and the carbonyl carbon.
6. Create the carbinolamine intermediate by forming a bond between the amine nitrogen and the aldehyde carbon.
7. Modify the structure to represent the iminium intermediate by removing water.
8. Position the indole nucleophile so that the C3 atom points toward the β -carbon of the iminium intermediate (approximately 3–3.5 Å).
9. Form the carbon–carbon bond corresponding to the Friedel–Crafts reaction.
10. Introduce a water molecule to simulate hydrolysis and regenerate the pAF residue.

7 Conclusion

The artificial enzyme LmrR pAF provides a powerful example of how unnatural amino acids can be used to introduce new catalytic mechanisms into protein scaffolds. Through iminium catalysis, the pAF residue activates aldehyde substrates and enables efficient Friedel–Crafts alkylation reactions with indole nucleophiles. Structural analysis of the enzyme reveals how the hydrophobic pocket at the LmrR dimer interface helps orient substrates and stabilize intermediates, allowing the protein to function as an effective catalytic platform.

COMMENTS

Hi Nayanika

I have the following questions to the structures:

reaction 1: The pAF looks "loose" inside the protein instead of being integrated into the peptide backbone - is that intentional? (see LmrR_pAF structure for reference: <https://www.rcsb.org/structure/6I8N> - that crystal structure has a molecule of MES buffer in it, that you can ignore, but pay attention to the pAF sidechains), also we have two pAF side chains per LmrR pocket, not just one, because it's a dimer (although we think only one of them is participating in the reaction at the same time. Connected to that - the position 15 (in natural LmrR that would be a V and in our system that's where the pAF is - it looks like the alpha helix just skips the position and forms the peptide bond over the gap where the pAF would be, thus the alpha-helix looks quite distorted there. See annotated screenshot below.

reaction 2: same comments as reaction 1.

reaction 3: same comments as reaction 1. Additionally, the product looks off. The pAF is already released in the structure, that means that the aldehyde should have reformed (the catalysis product is an aldehyde, not an alcohol, we only form the alcohol during the reductive workup after the reaction time has elapsed to arrest the distribution of product/substrate to analyze conversion). Furthermore in that case several atoms are not correctly hybridized. See annotation of my second screenshot.

If reaction 3 is not supposed to be the product but essentially step 2.2 in the reaction scheme I sent you last time, then following annotations apply instead (screenshot 3) in addition to the comments in reaction 1: